



PUMPS

A Complete Engineering Course

Classification • Centrifugal • Reciprocating • Surface • Submersible • Selection

Prepared by

Awet G. Nway

BSc Mechanical Engineering • Robotics & PLC Specialist

Founder, AYE Tech Hub

Engineering the Future through Technology & Knowledge



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01. What is a Pump?

Definition

A pump is a mechanical device used to increase the pressure energy of a fluid. It converts mechanical energy (from a prime mover such as an electric motor) into hydraulic energy in the form of velocity, pressure, or elevation of the fluid.

Key Functions of a Pump

- Raise fluid from a lower level to a higher level against gravity.
- Transfer fluid from one point to another through pipelines.
- Increase pressure of fluid in a system (e.g. hydraulic press, boiler feed).
- Maintain fluid circulation in closed systems (cooling, heating loops).
- Create vacuum or suction to draw fluid from deep sources.
- Provide accurate metering and dosing of chemicals or fluids.

Prime Movers Used with Pumps

Prime Mover	Power Source	Common Application
Electric Motor	Electrical energy	Most industrial & domestic pumps
Diesel / Petrol Engine	Fuel combustion	Remote sites, emergency pumps
Steam Turbine	Steam energy	Power plant boiler feed pumps
Hydraulic Motor	Hydraulic pressure	Mobile equipment pumps
Solar / Wind (via motor)	Renewable energy	Remote water supply systems

02. Classification of Pumps — Master Overview

All pumps fall into two broad categories. Surface and Submersible pumps are installation-based subcategories that exist within both main families — most commonly within Centrifugal (Dynamic) pumps.

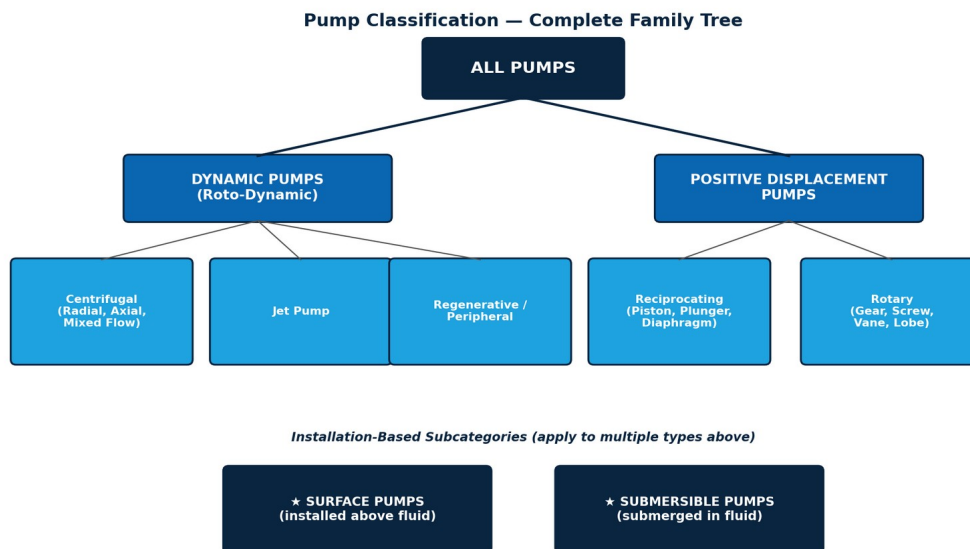


Figure 2.1 — Complete pump classification family tree showing dynamic and positive displacement branches.

1. Dynamic Pumps (Roto-Dynamic)

- **Centrifugal Pumps** — Volute / Radial flow, Axial flow, Mixed flow, Turbine / Diffuser.
- **Other Dynamic** — Jet pumps, Regenerative pumps.
- **By Installation** — ★ Surface Pumps ★ Submersible Pumps.

2. Positive Displacement Pumps

- **Reciprocating Pumps** — Piston pump, Plunger pump, Diaphragm pump.
- **Rotary Pumps** — Gear pump, Screw pump, Vane pump, Lobe pump.
- **By Installation** — ★ Surface mounted ★ Submersible (sewage).

Important Note on Installation Classification

Surface and Submersible are NOT separate pump types — they describe WHERE and HOW a pump is installed. A centrifugal pump can be designed as a surface pump OR a submersible pump. The same principle applies to some positive displacement pumps (e.g. submersible sewage pumps).

03. Centrifugal Pumps — Working Principle & Parts

A centrifugal pump works on the principle of centrifugal force. The rotating impeller imparts kinetic energy to the fluid, which is then converted to pressure energy in the volute casing — based on Bernoulli's Principle.

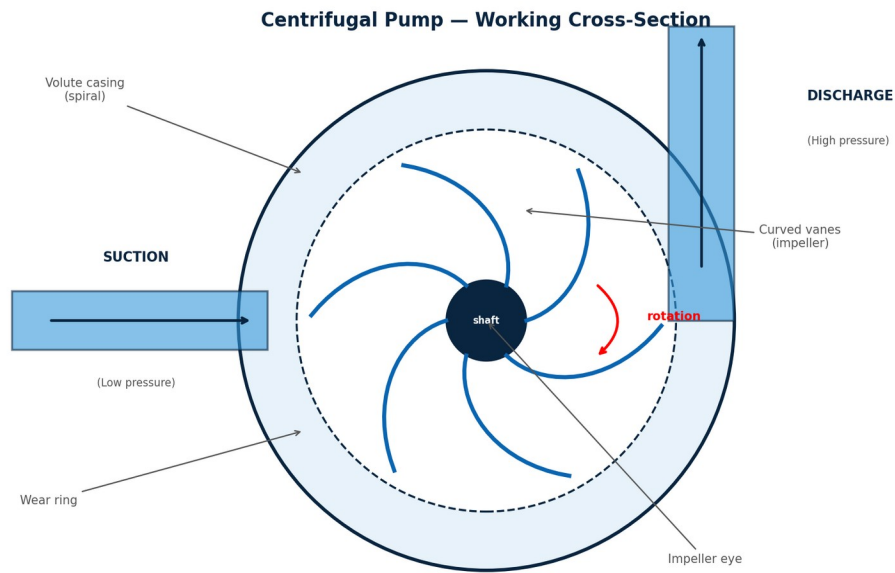


Figure 3.1 — Cross-section of a centrifugal pump showing impeller, volute casing, suction and discharge.

Step-by-Step Operation

- Fluid enters at the eye (centre) of the impeller through suction pipe.
- Rotating impeller throws fluid outward by centrifugal force.
- Fluid velocity increases from eye to impeller periphery.
- High-velocity fluid enters volute casing — area gradually increases.
- Increasing area → velocity decreases → kinetic energy becomes pressure energy.
- High-pressure fluid exits through discharge pipe.
- Low pressure at eye draws more fluid in — continuous flow established.

Basic Parts

Part	Function
Impeller	Rotating disc with curved vanes — main working part, imparts velocity to fluid.



Part	Function
Volute Casing	Spiral casing converting velocity to pressure, guides fluid to discharge.
Suction Pipe	Connects fluid source to pump inlet — fitted with strainer and foot valve.
Discharge Pipe	Carries pressurised fluid from pump to delivery point.
Shaft	Connects impeller to motor, transmits rotational energy.
Bearings	Support shaft and allow rotation with minimal friction.
Mechanical Seal	Prevents fluid leakage where shaft exits casing.
Wear Rings	Replaceable rings reducing recirculation leakage between impeller and casing.



04. Classification of Centrifugal Pumps

Centrifugal pumps are classified by several design and installation criteria. Understanding these helps you specify the right configuration for any application.

Based on Shaft Orientation & Installation

Type	Description
Horizontal	Shaft horizontal. Easy maintenance access. Most common industrial pump orientation.
Vertical	Shaft vertical. Space-saving. Used in sump pumps and wet pit installations.
★ Surface Pump	Pump installed ABOVE fluid level. Suction pipe draws fluid up. See Section 05 for full detail.
★ Submersible	Entire pump and motor submerged in fluid. Pushes fluid up. See Section 06 for full detail.

Based on Number of Stages

Type	Description
Single Stage	One impeller. Low to moderate head. Simple. Most common type for water supply.
Multi Stage	2+ impellers in series. Each adds pressure. For high head — boiler feed, RO, deep well.

Based on Type of Impeller

Type	Description
Closed Impeller	Vanes between two shrouds. Highest efficiency. For clean liquids only.
Semi-Open Impeller	One shroud. Handles mildly contaminated liquids. Moderate efficiency.
Open Impeller	No shrouds. For slurries and fibrous fluids. Lowest efficiency but easiest to clean.

Based on Number of Suctions

Type	Description
Single Suction	Fluid enters from one side of impeller only. Standard design.
Double Suction	Fluid enters both sides. Balances axial thrust. Used for very high flow rates.



Based on Flow Direction Through Impeller

Type	Description
Radial Flow	Fluid exits perpendicular to shaft. High head, low flow. Most common centrifugal type.
Axial Flow	Fluid exits parallel to shaft (propeller action). Very high flow, low head. Flood control.
Mixed Flow	Combination. Medium head and flow. Irrigation, cooling towers.

Based on Casing Design

Type	Description
Volute Casing	Spiral casing. Simple, converts velocity to pressure. Most common.
Diffuser Casing	Fixed vanes around impeller. High efficiency. Used in multistage pumps.
Vortex Casing	Large circular chamber. Handles solids and slurry well.



05. Surface Pumps

Definition

A surface pump (also called an above-ground pump) is a pump that is installed above the fluid source. It uses suction pressure created by the pump to draw fluid upward from a well, tank, or reservoir through a suction pipe, then discharges it under pressure to the delivery point. The motor and pump unit are always accessible above ground.

Working Principle

- Pump impeller rotates → creates low pressure (partial vacuum) at suction inlet.
- Atmospheric pressure on fluid surface pushes fluid up the suction pipe.
- Maximum theoretical suction lift = 10.3 m (atmospheric pressure limit).
- Practical suction lift limit = 6–8 m due to friction and NPSH requirements.
- Fluid enters pump, gains energy, exits under pressure through discharge pipe.
- Must be primed (filled with liquid) before starting to avoid dry running.

Types of Surface Pumps

Type	Description
Centrifugal Surface Pump	Most common surface pump. Impeller driven by motor above ground. Used for irrigation, water supply, HVAC. Requires priming. Cannot self-prime.
Self-Priming Centrifugal	Modified centrifugal pump with air-water separation chamber. Can prime itself without manual filling. Used where suction conditions are poor or intermittent.
Jet Pump (Shallow Well)	Uses a jet ejector on the suction side to create vacuum. Can lift water up to 8 m. Motor above ground, jet assembly near water. Common in domestic wells.
Jet Pump (Deep Well)	Two-pipe system — pressure pipe sends water down to ejector, return pipe brings water up. Can draw water from 25–30 m depth. Less efficient than submersible.
Peripheral / Regeneration	Multi-vane impeller in annular casing. High head at low flow. Self-priming. Used for booster pressure, small garden pumps.
End Suction Centrifugal	Suction at one end, discharge at top. Standard industrial surface pump configuration. Easy maintenance — back pull-out design common.
Split Casing Double Suction	Casing splits horizontally. Impeller with double entry. High flow, low NPSH. Used in large water supply and irrigation pumping stations.



Advantages

- Easy to inspect, maintain and repair.
- Lower purchase cost.
- Motor stays dry — longer life.
- Easy to monitor running conditions.
- No special waterproof motor needed.
- Simple installation above ground.
- Can be used with various pipe sizes.

Disadvantages

- Limited suction lift (max 6–8 m).
- Cannot pump from deep sources.
- Requires priming before start.
- Cavitation risk if suction too long.
- Not suitable for deep boreholes.
- Exposed to weather — needs housing.
- Audible noise during operation.

Applications of Surface Pumps

Application	Type Used	Why Surface Pump?
Domestic water supply (shallow well)	Jet pump / Self-priming	Easy installation, accessible motor
Agricultural irrigation	Centrifugal end-suction	High flow, low head, cheap
Swimming pool circulation	Centrifugal surface pump	Easy maintenance, dry motor
HVAC chilled/hot water circulation	Inline centrifugal	Accessible for servicing
Booster pump sets (buildings)	Multi-stage centrifugal	Pressure boosting from tank
Fire fighting pump set	End suction centrifugal	Reliable, accessible, testable
Industrial process transfer	End suction / split casing	High flow, accessible seals



06. Submersible Pumps

Definition

A submersible pump is a pump where the entire unit — motor and pump — is submerged inside the fluid being pumped. Unlike surface pumps that pull fluid up, submersible pumps push fluid upward, which is more efficient and eliminates suction lift limitations. The motor is hermetically sealed and rated IP68 (fully waterproof) to prevent fluid ingress.

Working Principle

- Entire pump + motor assembly lowered into fluid — always flooded and primed.
- Motor drives impeller(s) directly — no suction lift required.
- Fluid surrounding motor acts as coolant — motor stays cool even at full load.
- Pump pushes fluid up the delivery column pipe to surface — more efficient than pulling.
- Multiple impeller stages in series achieve high pressure for deep installations.
- Motor sealed with mechanical seals, O-rings, and moisture-proof windings.
- Cable entry sealed with waterproof gland — rated for continuous submersion.

Types of Submersible Pumps

Type	Description
Borehole / Deep Well Pump	Long, slim multi-stage centrifugal pump designed to fit inside narrow boreholes (4" to 12" diameter). Used for drinking water supply from deep aquifers — depth 10 m to 300 m+.
Sewage / Wastewater Pump	Handles raw sewage, solids, and fibrous materials. Open or vortex impeller. Installed in sump pits. Used in municipal sewage lifting stations and buildings.
Drainage / Dewatering Pump	Removes clean or slightly dirty water from flooded areas, excavations, basements. Semi-open impeller. Portable — can be moved between sites.
Sump Pump	Small submersible pump for automatic drainage of sump pits. Float switch activates when water level rises. Common in basements and car parks.
Slurry Submersible Pump	Handles abrasive slurries — mining, dredging, sand extraction. Hardened impeller and casing materials (hard iron, rubber lined).
Fountain / Pool Pump	Small decorative submersible pumps for fountains, ponds, aquariums. Low head, low flow. Quiet operation.
Oil / Petroleum Pump	Submersible Electric Submersible Pump (ESP) used inside oil wells. Multi-stage centrifugal, handles oil-water-gas mixtures. Depths up to 3000 m.



Type	Description
Condensate Pump	Small submersible or semi-submersible for removing condensate from air conditioning units and boiler systems.

Advantages

- No priming required — always flooded.
- No suction lift limitation.
- Can pump from very deep sources (300 m+).
- Quiet operation — motor underwater.
- Efficient — pushes fluid (not pulls).
- Fluid cools motor — no overheating.
- No need for pump house above ground.
- Self-contained unit — compact installation.

Disadvantages

- Difficult to inspect — must be extracted.
- Higher purchase cost than surface pump.
- Motor failure requires full extraction.
- Seal failure allows water into motor.
- Cable damage risk during installation.
- Cannot run dry — must stay submerged.
- Requires special waterproof motor.
- Harder to monitor running conditions.

Applications of Submersible Pumps

Application	Type Used	Depth / Condition
Deep borehole water supply	Multi-stage borehole pump	50–300 m depth
Sewage lifting station	Sewage submersible pump	Sump pit 2–10 m
Basement flood drainage	Sump / drainage pump	Automatic float switch
Mining dewatering	Slurry submersible pump	Underground, abrasive water
Oil well production (ESP)	Electric submersible pump	500–3000 m depth
Fountain and pond	Small submersible pump	Decorative, shallow
Offshore platform drainage	Submersible sewage pump	Marine grade IP68



Application	Type Used	Depth / Condition
Construction site dewatering	Portable drainage pump	Temporary, moved daily

07. Surface vs Submersible Pump — Full Comparison

Below is a complete side-by-side comparison to guide selection between surface and submersible pump configurations.

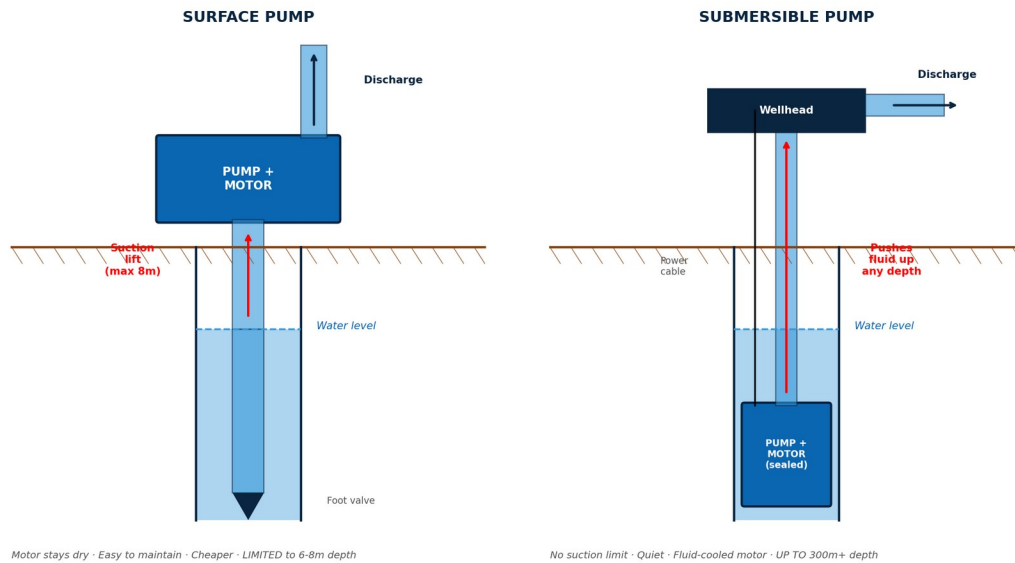


Figure 7.1 — Surface pump (left) draws fluid up through suction pipe. Submersible pump (right) pushes fluid up from inside the well.

Property	Surface Pump	Submersible Pump
Installation location	Above the fluid	Submerged in fluid
Priming	Required before start	Not required — always flooded
Suction lift	Max 6–8 m practical	No limit — pushes fluid up
Depth capability	Up to ~8 m suction	Up to 300 m+ depth
Efficiency	Moderate (suction losses)	Higher (pushes fluid)
Motor cooling	Air cooled	Fluid cooled
Noise level	Audible above ground	Silent (motor underwater)
Maintenance access	Easy — motor above ground	Must extract from fluid
Motor protection	Standard motor OK	Needs IP68 sealed motor
Cavitation risk	Higher (suction side)	Very low (always flooded)



Property	Surface Pump	Submersible Pump
Dry running risk	Yes — needs priming check	Yes — must stay submerged
Installation cost	Lower	Higher
Motor replacement	Simple on-site	Full extraction required
Best for shallow water	Yes — up to 8 m	Possible but overkill
Best for deep water	No — limited suction	Yes — ideal
Suitable for sewage	No — solids block suction	Yes — vortex impeller type
Borehole use	Jet pump only (limited)	Yes — primary choice
Typical applications	Irrigation, HVAC, domestic, fire pump	Boreholes, sewage, drainage, oil wells

Quick Selection Rule

If water source is within 6–8 m depth and easy access for maintenance is needed → choose Surface Pump. If water source is deep (>8 m), submerged in a pit, or requires quiet/compact installation → choose Submersible Pump.

08. Reciprocating Pumps — Working & Components

Definition

A reciprocating pump is a positive displacement pump in which a piston or plunger moves back and forth inside a cylinder to displace a fixed volume of fluid per stroke. It delivers accurate volume regardless of pressure — ideal for high pressure and metering applications.

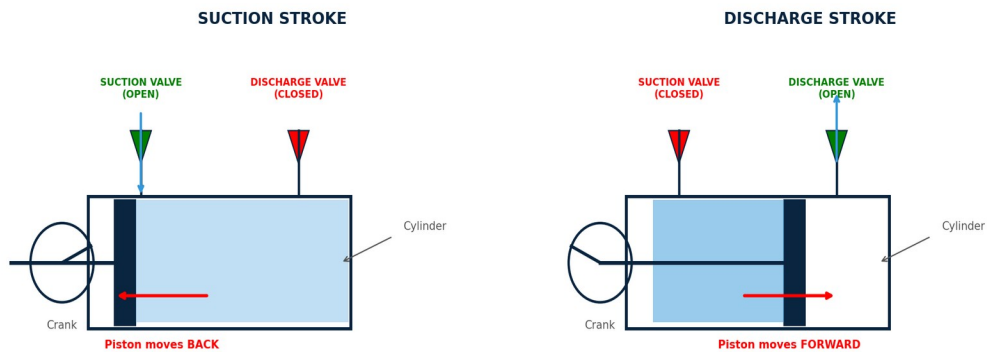


Figure 8.1 — The two strokes of a reciprocating pump: suction (left) draws fluid in, discharge (right) pushes it out.

Step-by-Step Working

- **Suction Stroke** — Piston moves backward → cylinder volume increases → pressure drops → suction valve opens → fluid drawn into cylinder.
- **Discharge Stroke** — Piston moves forward → volume decreases → pressure rises → suction valve closes → discharge valve opens → fluid forced out under pressure.
- **Valve Action** — Non-return (check) valves on both suction and discharge — allow one-direction flow only, prevent backflow.
- **Continuous Flow** — Double-acting or multi-cylinder designs ensure near-continuous discharge — while one side discharges, the other is on suction stroke.

Key Components

Component	Function
Piston / Plunger	Reciprocating element. Piston = close-fit with rings. Plunger = rod-type, higher pressure.
Cylinder	Houses piston. Must withstand full operating pressure.
Suction Valve	Non-return valve on inlet — opens during suction stroke only.



Component	Function
Discharge Valve	Non-return valve on outlet — opens during discharge stroke only.
Crank & Connecting Rod	Converts rotary motor motion to reciprocating piston motion.
Air Vessel	Fitted on suction and discharge. Reduces pulsation and water hammer.
Packing / Seals	Prevent fluid leakage past piston along cylinder wall.



09. Classification of Reciprocating Pumps

Reciprocating pumps are classified by piston action, number of cylinders, the displacing element, drive mechanism, and installation.

Based on Piston Action

Type	Description
Single-Acting	Discharge on forward stroke only. One set of valves. Simple but pulsating flow.
Double-Acting	Discharge on both forward and return strokes. Smoother flow, higher output.

Based on Number of Cylinders

Type	Description
Simplex (1 cyl)	One cylinder. High pulsation. Low flow. Simple construction.
Duplex (2 cyl)	Two cylinders 90° apart. Reduced pulsation. Common in industry.
Triplex (3 cyl)	Three cylinders 120° apart. Smooth flow. Common in high-pressure wash, oilfield.
Quintuplex (5 cyl)	Very smooth flow. High-pressure oilfield injection and metering.

Based on Displacing Element

Type	Description
Piston Pump	Piston fits tightly in cylinder with rings. Lower pressure. General industrial use.
Plunger Pump	Solid plunger — sealing at cylinder end. Very high pressure (up to 1000+ bar).
Diaphragm Pump	Flexible membrane replaces piston. No dynamic seals. Handles corrosive, toxic, slurry fluids.

Based on Drive Mechanism

Type	Description
Crank Driven	Standard crank and connecting rod. Most common.
Cam Driven	Cam profile controls stroke. Metering pumps.
Hydraulically Actuated	Hydraulic oil drives plunger. High pressure metering pumps.



Based on Installation

Type	Description
Surface Mounted	All reciprocating pumps are normally surface-mounted. Motor and pump above ground.
Submersible Sewage (Special Type)	Some diaphragm and small piston pumps are submersible for sewage/slurry duty. Not common.

10. Centrifugal vs Reciprocating Pump — Full Comparison

Below is a complete 15-property comparison between the two most important pump families.

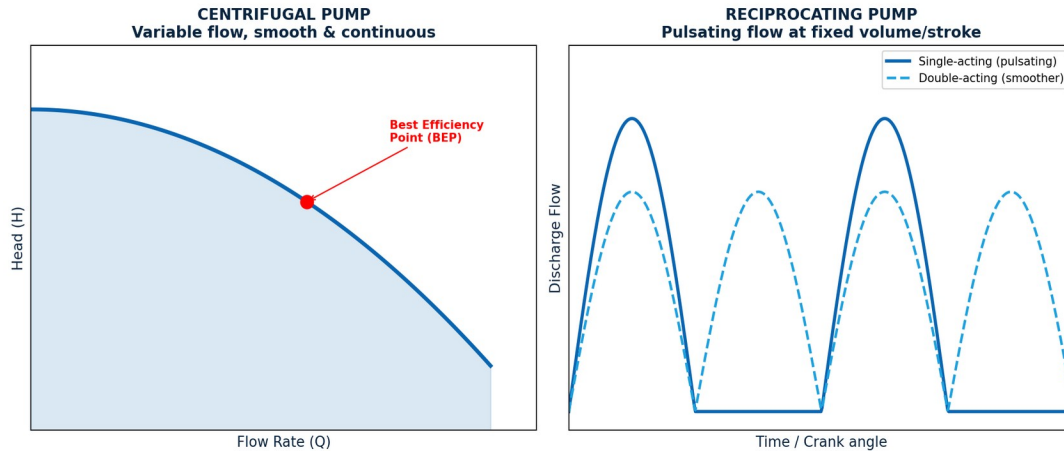


Figure 10.1 — Flow characteristics: centrifugal pumps give a smooth Q-H curve; reciprocating pumps produce pulsating fixed-volume discharge.

Property	Centrifugal Pump	Reciprocating Pump
Flow Rate	High flow, low-medium pressure	Low flow, high pressure
Pressure Range	Up to ~100 bar (multistage)	Up to 1000+ bar
Flow Type	Continuous, smooth	Pulsating (needs air vessel)
Self-Priming	No (most types)	Yes — can prime itself
Speed	High (1450–3000 RPM)	Low (50–300 RPM)
Efficiency	High near BEP (~80%)	High overall (~90%)
Viscous Fluids	Not suitable (>200 cP)	Excellent for viscous fluids
Size and Weight	Compact and lightweight	Large and heavy
Maintenance	Simple — few moving parts	Complex — valves and seals
Deadheading Risk	Safe — no pressure buildup	DANGEROUS — relief valve needed
Metering Accuracy	Poor — flow varies with pressure	Excellent — fixed volume/stroke
Starting	Start unloaded (close valve)	Can start under load
Noise & Vibration	Low	Higher (reciprocating motion)



Property	Centrifugal Pump	Reciprocating Pump
Initial Cost	Low	High
Best Applications	Water supply, HVAC, irrigation	High pressure, metering, oilfield

11. Pump Selection Guidelines

Choosing the right pump for any application starts with understanding the system requirements and matching them to pump capabilities. Below is the standard engineering approach.

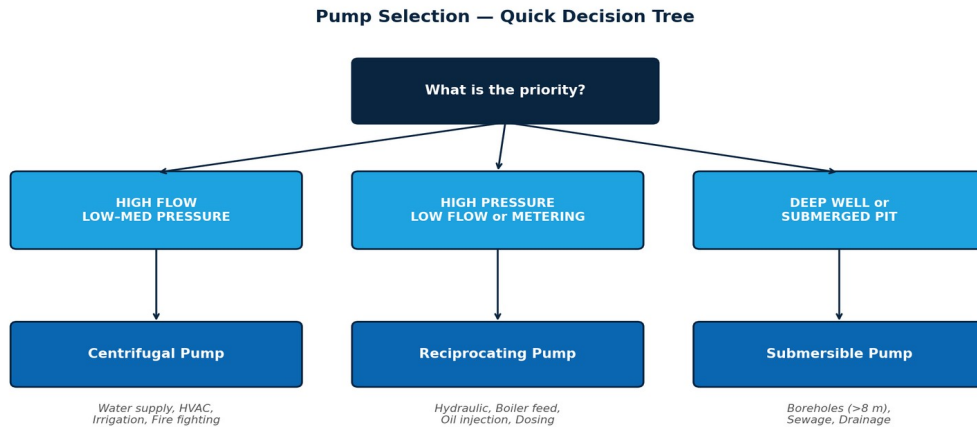


Figure 11.1 — Pump selection decision tree based on flow, pressure and installation priorities.

General Pump Selection Checklist

Factor	What to Determine
Flow Rate (Q)	Calculate required flow in litres/min or m ³ /hr at operating conditions.
Total Head (H)	Static head + friction losses + velocity head + minor losses.
Fluid Properties	Density, viscosity, temperature, corrosiveness, solids content, vapour pressure.
NPSH Available	Ensure NPSHa > NPSHr to prevent cavitation (especially surface pumps).
Installation Type	Surface or submersible? Horizontal or vertical shaft? Indoor or outdoor?
Duty Type	Continuous, intermittent, standby? Single or duty/standby pair?
Material Selection	Cast iron, stainless steel, bronze, rubber-lined — match to fluid.
Motor Power	Calculate based on pump power at rated duty point + efficiency + margin.
Standards	API 610, ISO 5199, EN 733 — select applicable standard for application.



Choose CENTRIFUGAL — SURFACE PUMP when:

- Water source is within 6–8 m suction depth.
- High flow rate needed at moderate pressure — irrigation, HVAC, firefighting.
- Easy maintenance access is a priority.
- Installation above ground is possible.
- Budget is limited — lowest initial cost option.

Choose CENTRIFUGAL — SUBMERSIBLE PUMP when:

- Water source is deeper than 8 m (borehole, deep well).
- Pump must operate in a submerged pit (sewage, drainage, sump).
- Quiet operation is required — residential, indoor installation.
- No space for pump house above ground.
- Continuous automatic operation with float switch control.

Choose RECIPROCATING PUMP when:

- Very high pressure delivery required (100–1000+ bar).
- Accurate metering and dosing of fluid is critical.
- Fluid is viscous, slurry, or chemically aggressive.
- Self-priming capability is needed without modification.
- Low flow at high pressure is the operating condition.



12. Key Formulas & Calculations

Below are the essential engineering equations used in pump sizing, selection, and analysis. Memorize these for any pump-related interview or design calculation.

Flow & Continuity

- **Flow Rate:** $Q = A \times v$ (m^3/s) where A = pipe area, v = fluid velocity.
- **Continuity:** $A_1 \times v_1 = A_2 \times v_2$ (flow conserved in pipe).
- **Pipe Velocity:** $v = 4Q / (\pi \times D^2)$ (m/s).

Head Calculations

- **Total Head:** $H_{\text{total}} = H_{\text{static}} + H_{\text{friction}} + H_{\text{velocity}} + H_{\text{minor}}$ (m).
- **Static Head:** $H_s = Z_{\text{discharge}} - Z_{\text{suction}}$ (m).
- **Friction Head (Darcy-Weisbach):** $H_f = f \times (L/D) \times v^2 / 2g$.
- **Velocity Head:** $H_v = v^2 / (2g)$ where $g = 9.81 \text{ m/s}^2$.

Power & Efficiency

- **Hydraulic Power:** $P = \rho \times g \times Q \times H$ (Watts) where ρ = fluid density kg/m^3 .
- **Shaft Power:** $P_{\text{shaft}} = P_{\text{hydraulic}} / \eta_{\text{pump}}$.
- **Motor Power:** $P_{\text{motor}} = P_{\text{shaft}} / \eta_{\text{motor}}$.
- **Overall Efficiency:** $\eta = \eta_{\text{pump}} \times \eta_{\text{motor}} \times \eta_{\text{coupling}}$.

Reciprocating Pump Discharge

- **Single-Acting:** $Q = A \times L \times N / 60$ (m^3/s) where A = bore area, L = stroke, N = RPM.
- **Double-Acting:** $Q = (2A - a) \times L \times N / 60$ where a = rod area.
- **Volumetric Efficiency:** $\eta_{\text{vol}} = Q_{\text{actual}} / Q_{\text{theoretical}}$ (90–98% typical).

Affinity Laws — Centrifugal

- **Flow:** $Q_2/Q_1 = N_2/N_1$ (proportional to speed).
- **Head:** $H_2/H_1 = (N_2/N_1)^2$ (proportional to speed squared).
- **Power:** $P_2/P_1 = (N_2/N_1)^3$ (proportional to speed cubed).
- **Energy saving at 80% speed:** $P = 0.8^3 = 51.2\%$ of full-speed power.

NPSH — Cavitation Prevention

- **NPSH Available:** $\text{NPSHa} = (P_{\text{atm}} - P_{\text{vapour}}) / \rho g + Z_s - h_{\text{fs}}$ (m).
- **Safe Operation:** NPSHa must be greater than $\text{NPSHr} + 0.5 \text{ m}$ safety margin.
- **Suction Lift Limit:** Max practical suction = 6–8 m for surface pumps.
- **Submersible pumps:** NPSHa always sufficient — pump is submerged.





13. Real-World Applications

Each pump family has a natural home in industry. The map below shows where each type dominates and why.

Water Supply & Utilities

- Municipal water supply — surface centrifugal pumps at treatment plants.
- Deep borehole drinking water — submersible multi-stage pumps.
- Domestic household water — surface jet pump or submersible borehole pump.
- Desalination (RO) — high-pressure reciprocating / centrifugal pumps.

Sewage & Drainage

- Sewage lifting stations — submersible sewage pumps with vortex impeller.
- Building basement drainage — automatic submersible sump pump.
- Construction site dewatering — portable submersible drainage pumps.
- Storm water management — large axial flow submersible pumps.

Agriculture & Irrigation

- Field irrigation — surface centrifugal pumps from rivers/canals.
- Deep well irrigation — submersible borehole pumps for groundwater.
- Chemical fertigation dosing — reciprocating metering pumps.
- Livestock water supply — surface self-priming centrifugal pumps.

Oil & Gas

- Oil well production — Electric Submersible Pumps (ESP) at 500–3000 m depth.
- Water injection — high-pressure triplex/quintuplex reciprocating pumps.
- Pipeline booster stations — large surface centrifugal pumps.
- Chemical injection — precision reciprocating metering pumps.

Power Generation

- Boiler feed — high-pressure multistage surface centrifugal pumps.
- Cooling water — large surface centrifugal pumps (double suction).
- Condensate extraction — small surface or submersible centrifugal.
- Chemical dosing to boiler water — reciprocating metering pumps.

Building Services & HVAC

- Hot/chilled water circulation — surface inline centrifugal pumps.
- Fire fighting system — surface end-suction centrifugal pump sets.



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- Pressure boosting — surface multistage centrifugal booster sets.
 - Swimming pool — surface centrifugal or small submersible.



14. Final Words from AYE Tech Hub

Pumps are everywhere — from the water flowing into your home, to the oil rising from kilometres beneath the earth, to the boilers powering entire cities. Understanding how they work, how they are classified, and how to select the right one for the job is a core competency for every mechanical, electrical, and industrial engineer.

This course covered the two great families of pumps — Dynamic and Positive Displacement — and the two main installation arrangements — Surface and Submersible. With the working principles, classifications, comparison tables, formulas, and applications in this guide, you have a complete reference to confidently specify, install, troubleshoot, and discuss any pump system.

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